



Service Bulletin

Service Bulletin Number: 4383578

Released Date: 28-jul-2014

Spark Plug Reuse Guidelines

Spark Plug Reuse Guidelines

This document provides guidelines for replacement and reuse of spark plugs. Use this information in conjunction with the content provided in the corresponding service manual.



13300030



Spark Plugs - New

Appearance

Clean, without any signs of use.

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13300033

Spark Plug - Good Used Plug

Appearance	Clean, without any defects; minor signs of use.
Action	Reuse



13300034

Spark Plug - Acceptable Deposits

Appearance	Slightly discolored.
Cause	Deposits from normal combustion.
Action	Reuse

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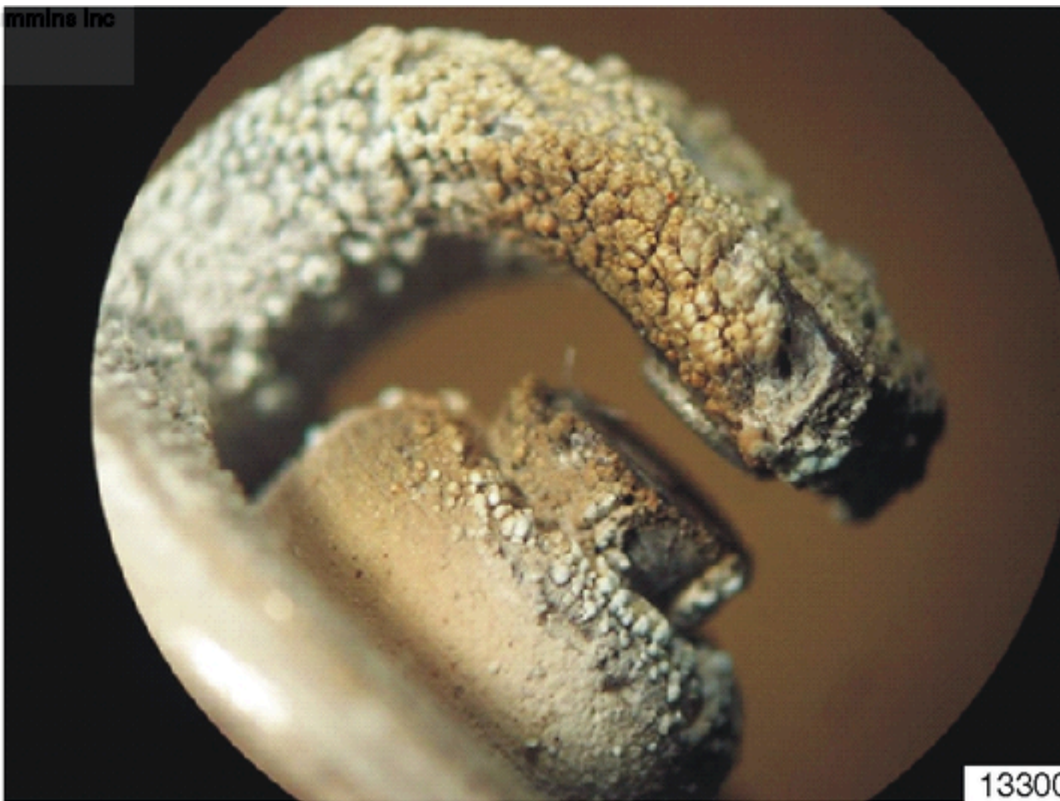
13300035

Spark Plug - Oil Deposits

Appearance	Coated with black, oily sludge.
Cause	Oil Deposits, leaking valve stem seals, piston malfunction, piston ring malfunction.
Action	Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Investigate possible causes of internal oil leaks and complaints of oil consumption. Replace all spark plugs with the appearance of the above deposits.



13300036



13300037

Spark Plug - Unacceptable Deposits

Appearance	Exceedingly discolored, or coated with ash.
Cause	Deposits from oil consumption, coolant consumption, or poor fuel quality.

Spark Plug - Unacceptable Deposits

Action

Check for excessive coolant consumption.
Check for excessive oil consumption.
Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Replace all spark plugs with the appearance of the above deposits.

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13300038

Spark Plug - Dry Fouling

Appearance

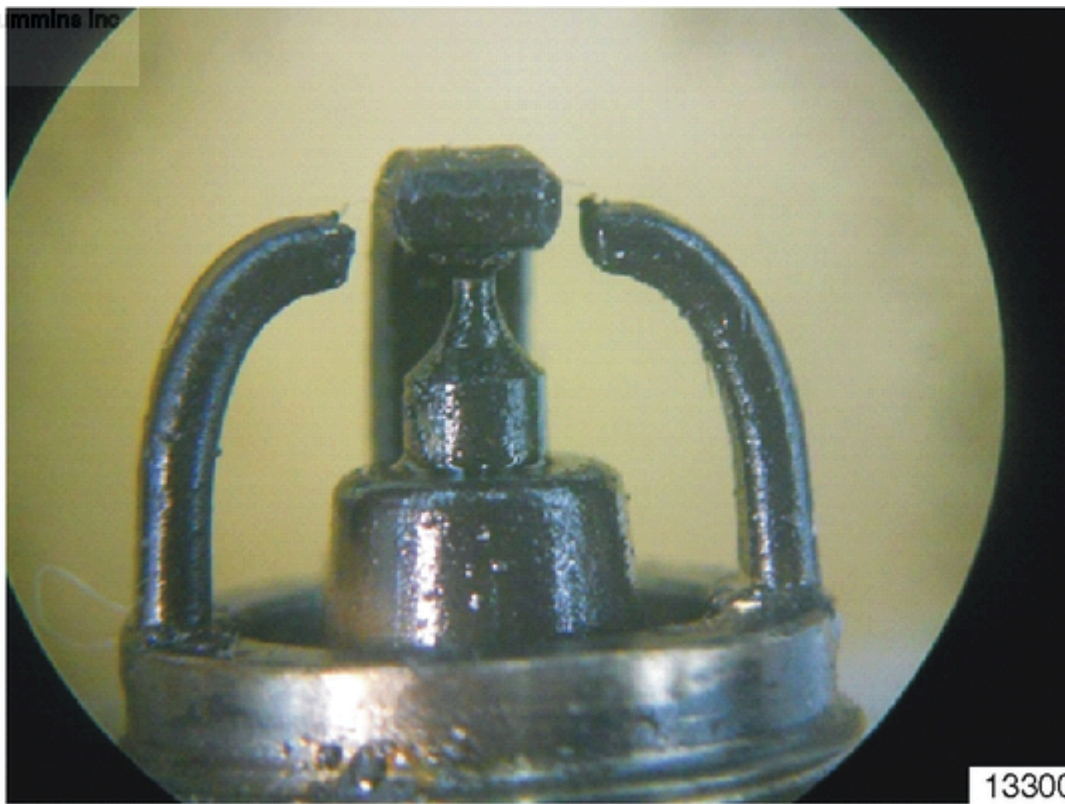
Dry, black deposits on electrodes and core nose.

Cause

Running rich for a prolonged period of time.

Action

Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Investigate possible causes of rich running condition. Replace all spark plugs with the appearance of the above deposits.



13300039

Spark Plug - Wet Fouling

Appearance	Wet, black deposits on electrodes and core nose.
Cause	Prolonged misfire.
Action	Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Investigate possible causes of misfire condition. Replace all spark plugs with the appearance of the above deposits.

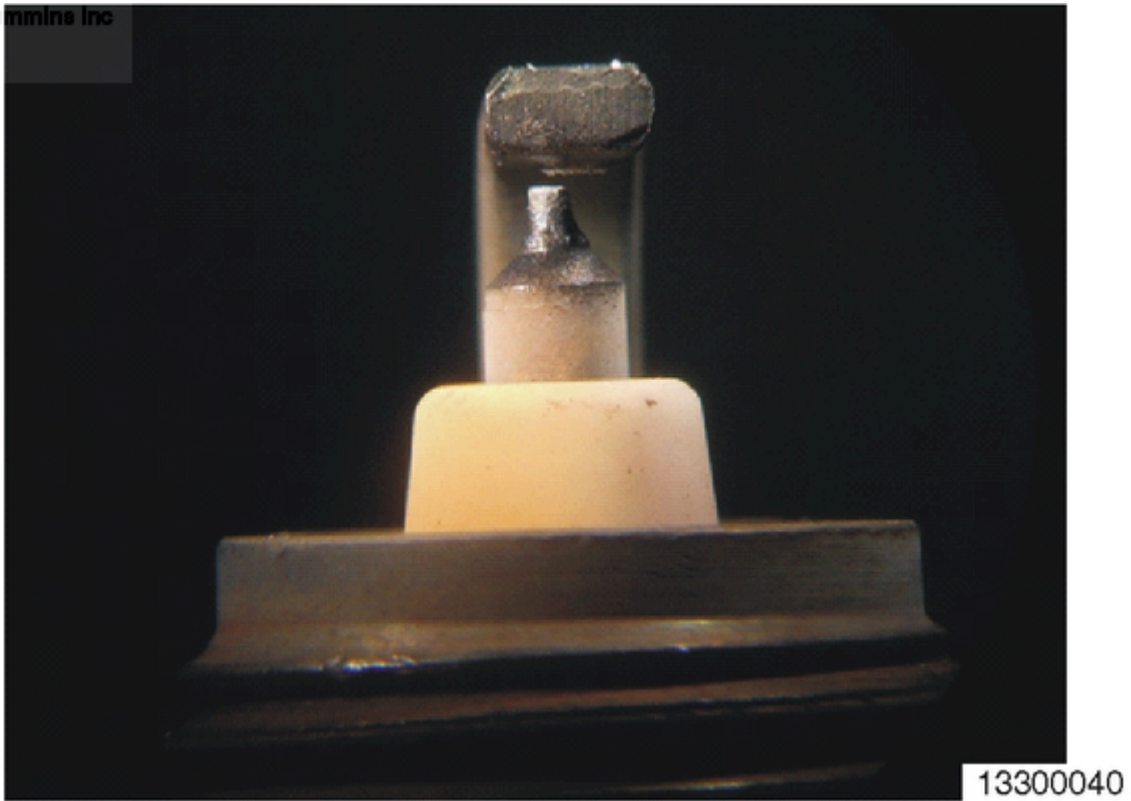


Figure 1

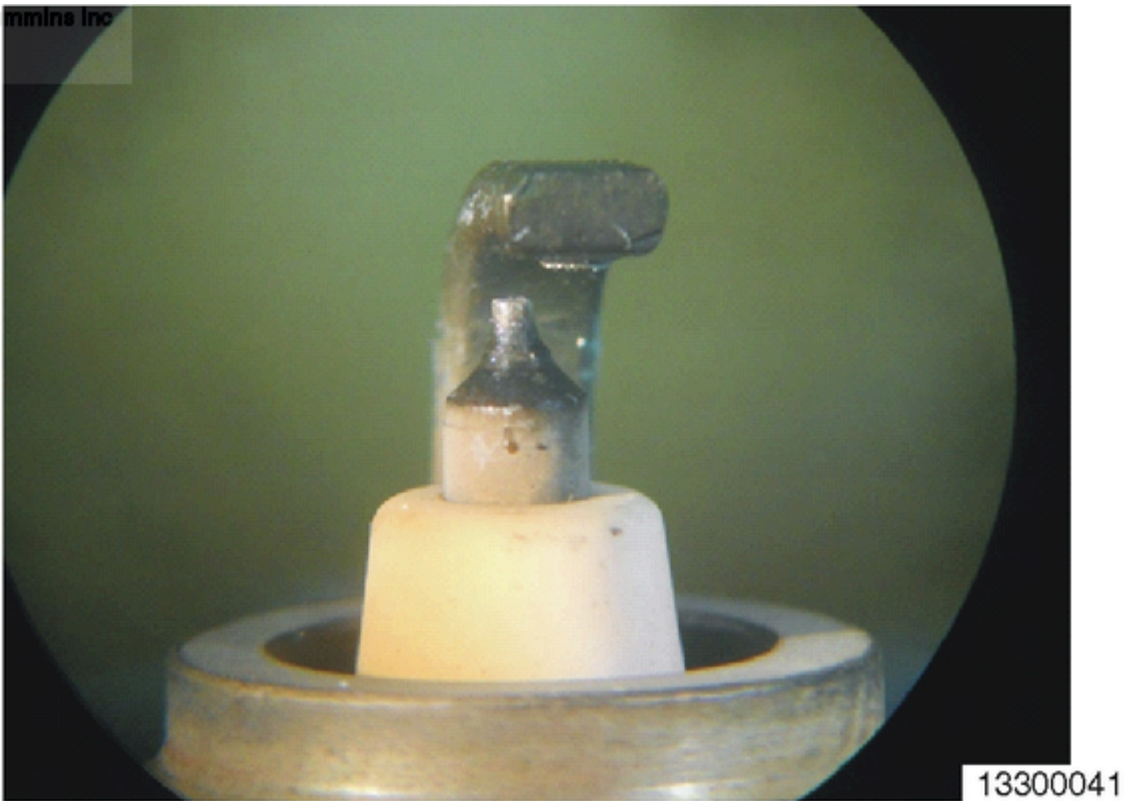


Figure 2

Spark Plug - Misaligned Electrode

Appearance

Precious metal pads do **not** overlap at all.
Figure 1 - Meets reuse guidelines. Figure 2 - Requires replacement.

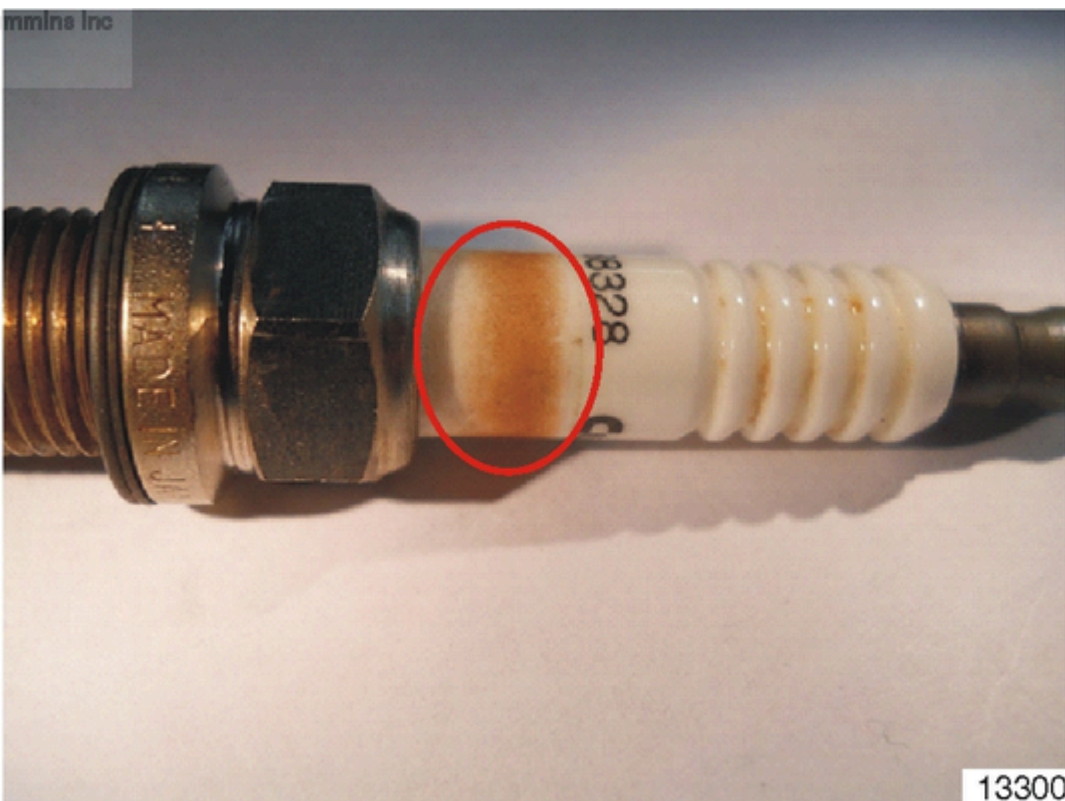
Cause

Mishandling.

Spark Plug - Misaligned Electrode

Action

Replace spark plug if metal pads do **not** overlap.



Spark Plug - Corona Staining	
Appearance	Normal brown discoloration of the insulator.
Cause	Electrically charged particles or dirt.
Action	Reuse

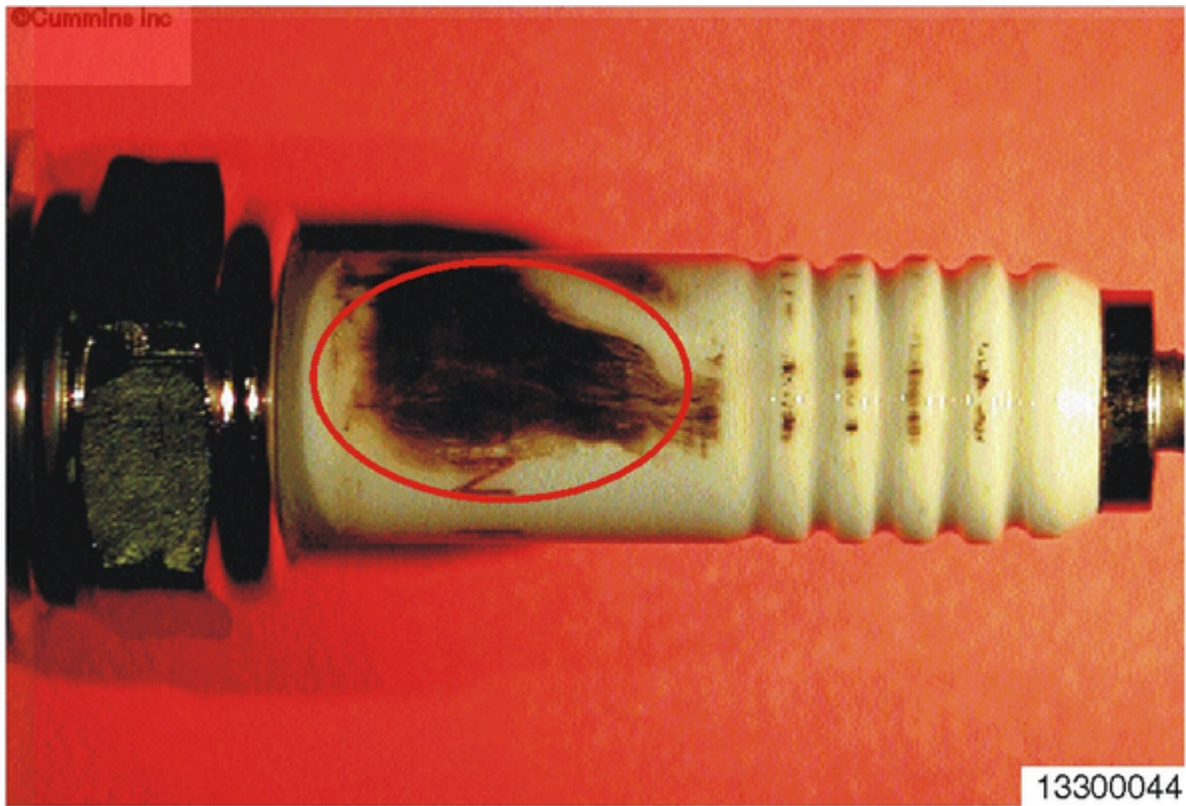


Figure 1

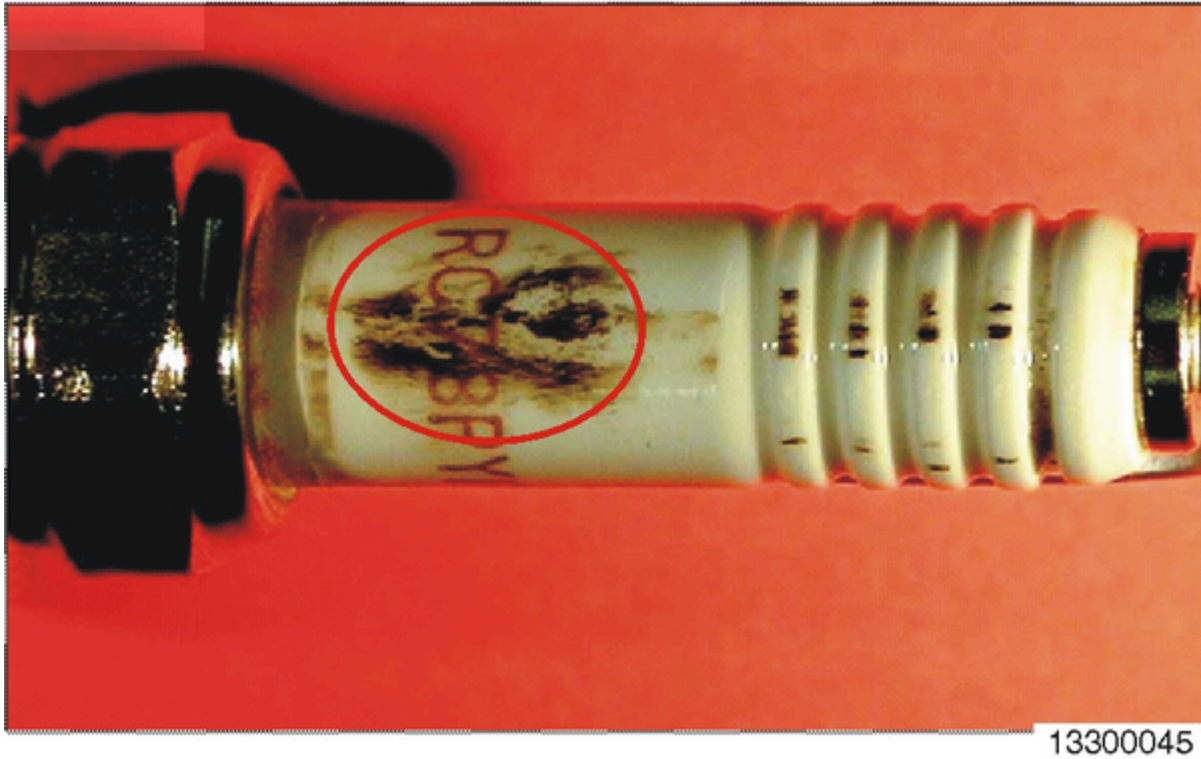


Figure 2

Spark Plug - Flashover

Appearance	Black or brown streaks on the insulator; Figure 1 - Extreme flashover, Figure 2 - Moderate flashover.
Cause	Contamination, damaged or worn spark plug boot.
Action	Replace the spark plug.



13300046

Spark Plug - Gasket Leak/Internal Seal Malfunction

Appearance	Burnt oil around insulator.
Cause	Over-torqued.
Action	Replace the spark plug only if the insulator rotates relative to the metal shell or shows heat damage.



13300047

Figure 1



13300048

Figure 2

Spark Plug - Closed Gap

Appearance

Ground electrode and center electrode touching; Figure 1 - Closed gap, Figure 2 - Meets reuse guidelines.

Cause

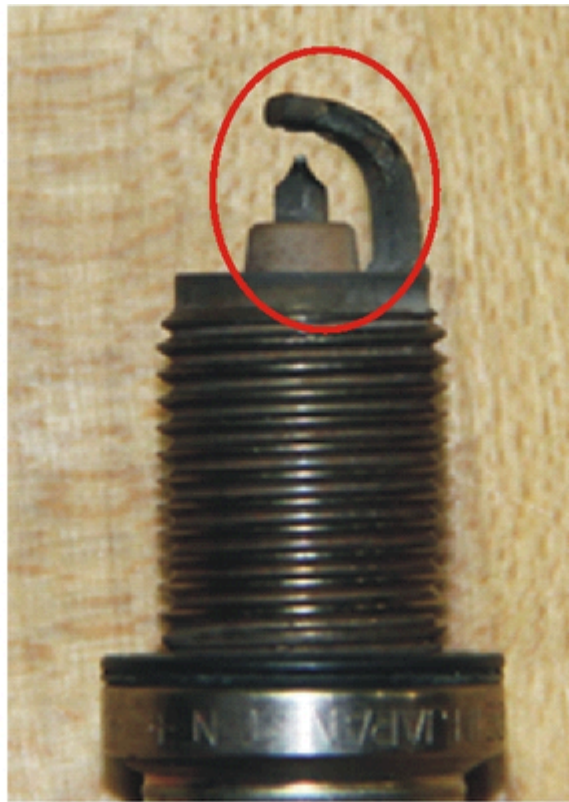
Mishandling or knock.

Spark Plug - Closed Gap

Action

Check for possible engine knock. Replace spark plug.

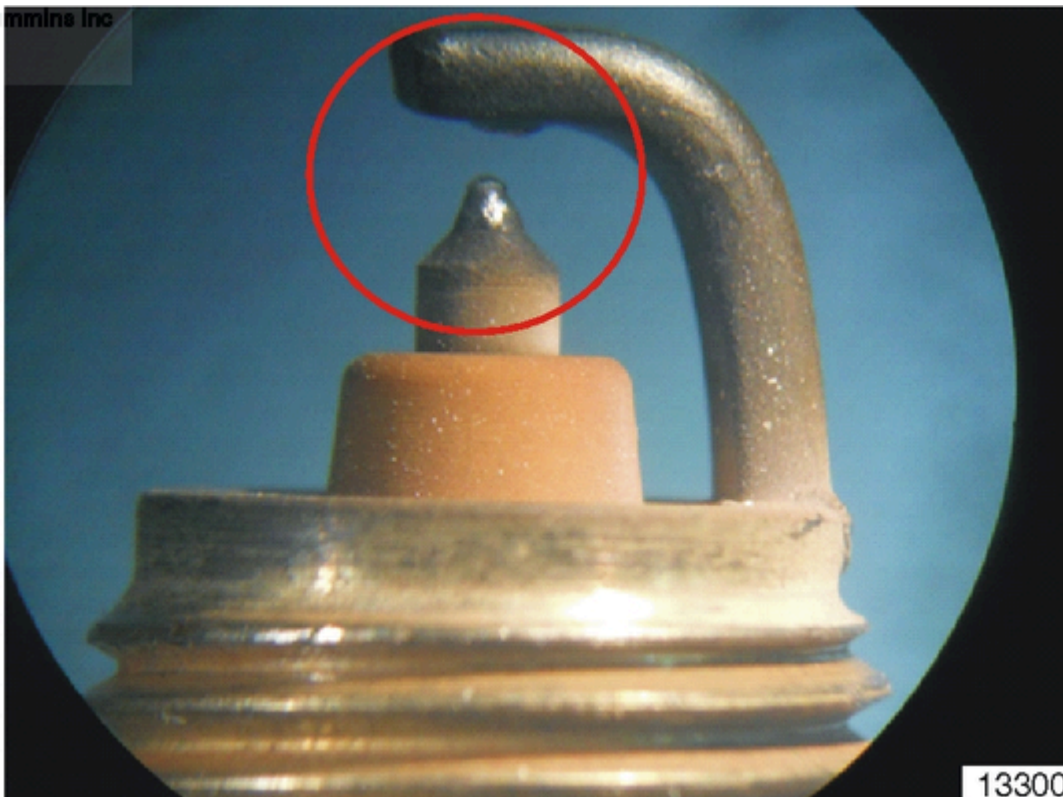
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Figure 1

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Figure 1



13300051

Figure 2

End of Life Spark Plug - Excessive Gap

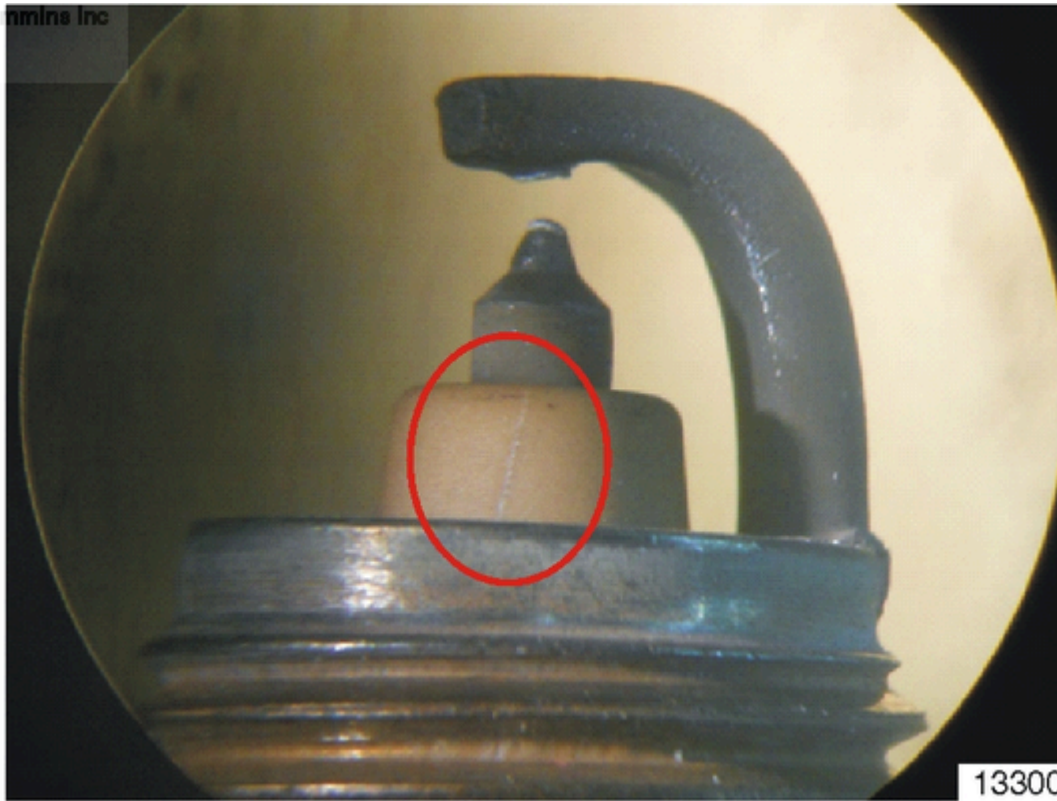
Appearance	Gap between the center electrode and ground electrode are out of specification. Figures 1 - Excessive gap. Figure 2 - Meets reuse guidelines.
Cause	Spark plug at or beyond its end of life.
Action	Replace all the spark plugs.



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13300054



13300052

Spark Plug - Core Nose Damage

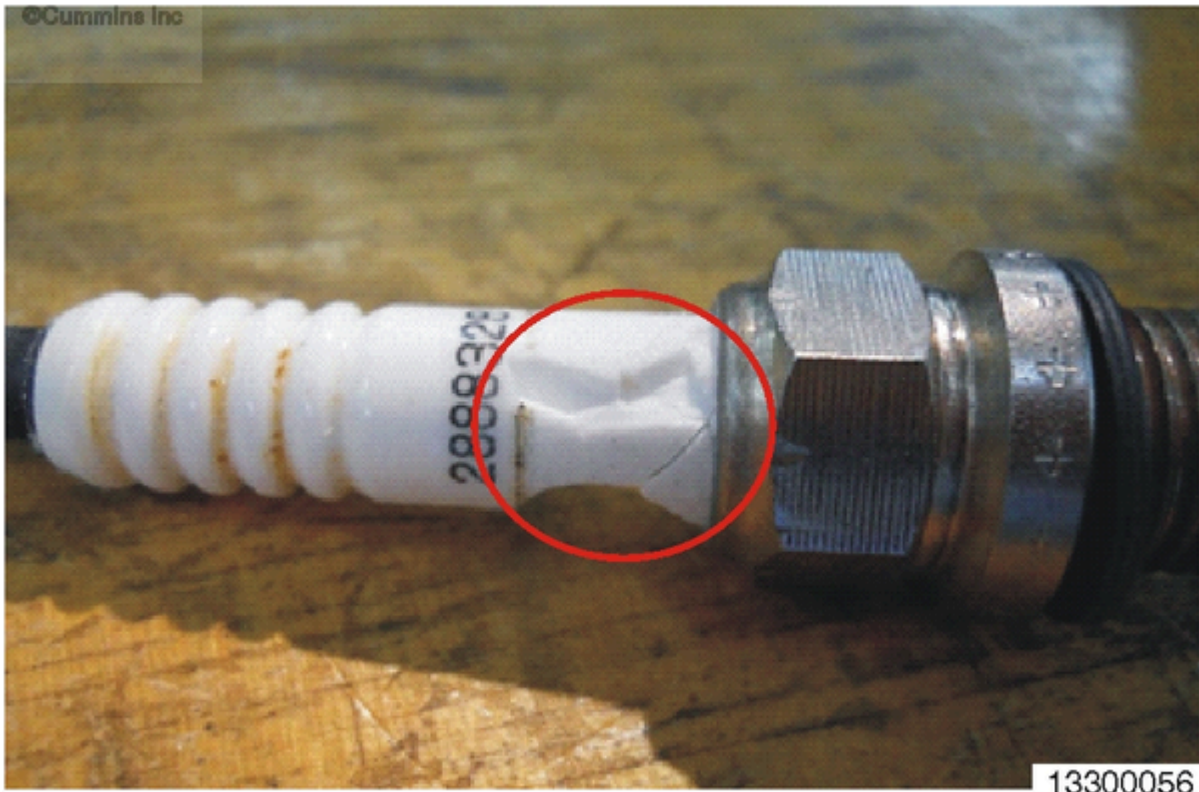
Appearance	Cracked, chipped, or missing core nose ceramic.
Cause	Mishandling, knock.
Action	Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Replace all spark plugs with a cracked core nose.



13300055

Spark Plug - Melted Ground Electrode

Appearance	Ground electrode melted and/or partially missing.
Cause	Excessive temperatures or knock.
Action	Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Check for in-cylinder damage or causes of knock. Replace all spark plugs with a melted ground electrode.



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13300057

Spark Plug - Cracked Insulator

Appearance

Cracked insulator.

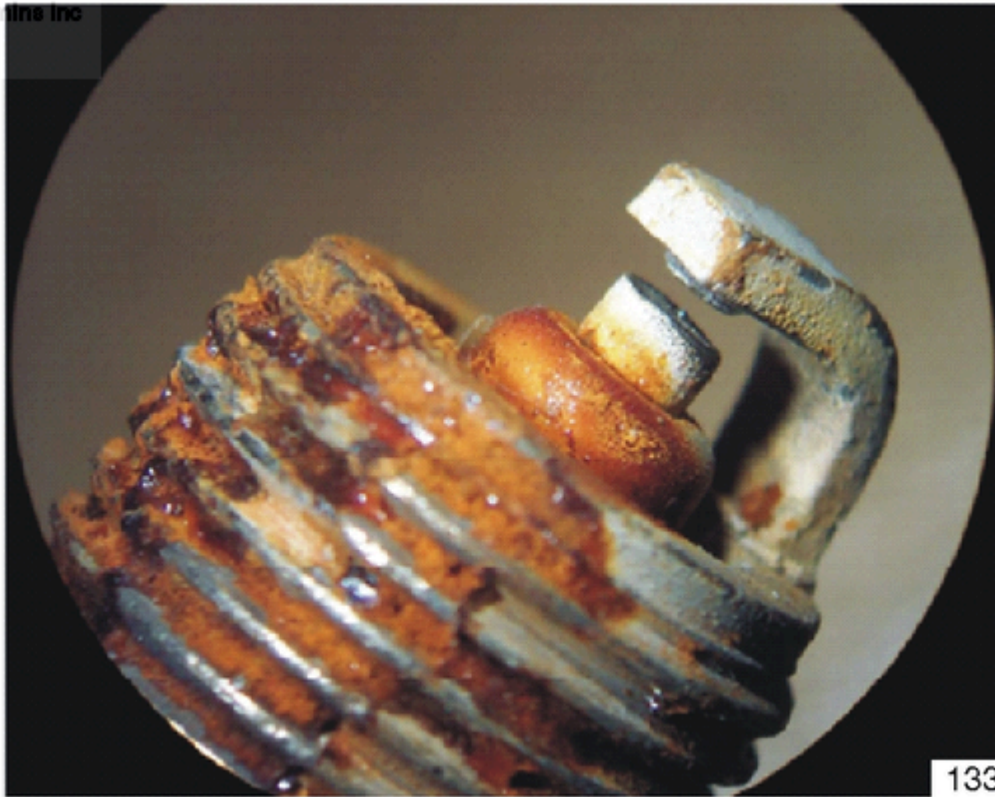
Cause

Mishandling or over-torqued.

Spark Plug - Cracked Insulator

Action

Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Replace all spark plugs with cracked insulators.

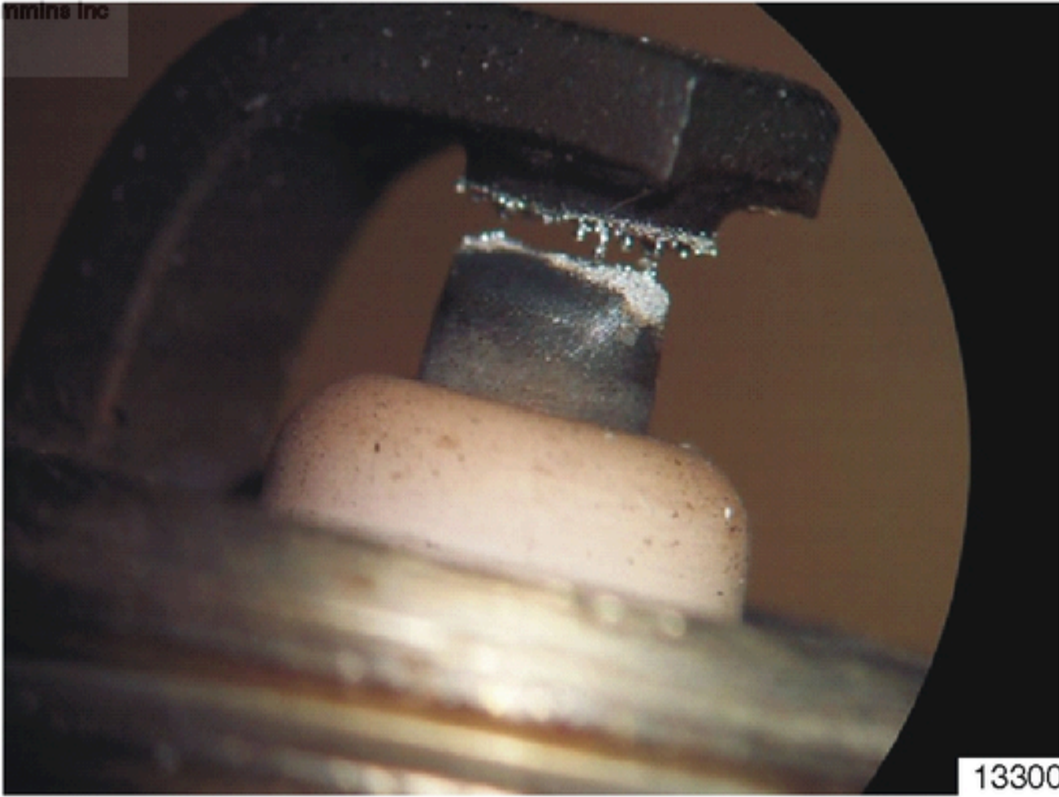




13300060

Spark Plug - Corrosion

Appearance	Discolored and/or rusted.
Cause	Moisture in the spark plug well and/or leaking seal or boot.
Action	Check the seal on ignition coil and spark plug boot. Investigate for the source of moisture. Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Replace all spark plugs with corrosion that will not wipe off.



13300061



13300062

Spark Plug - Beading

Appearance

Small beads on the electrodes, bridging the gap.

Cause

Excessive temperatures and/or fuel which does **not** meet Cummins® specifications.

Spark Plug - Beading

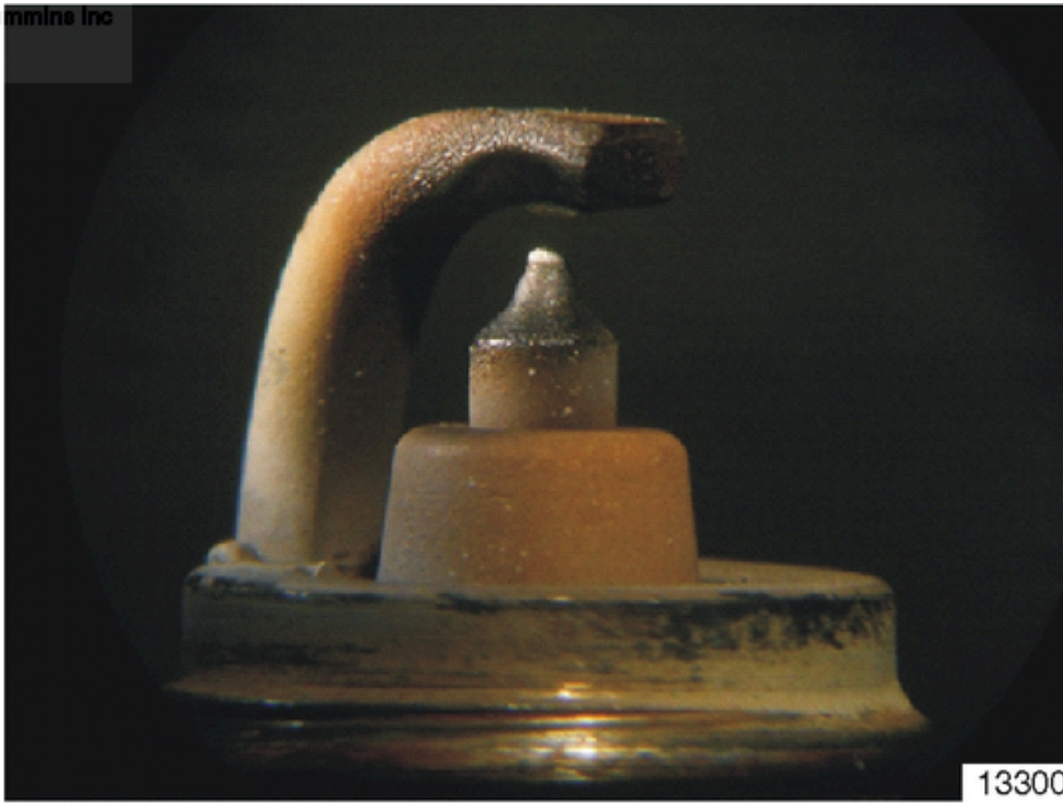
Action

Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Replace spark plug **only** if the gap is out of specification. Verify fuel quality and investigate source of excessive in-cylinder temperature.

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13300063

**Spark Plug - Uneven Wear on Ground Electrode**

Appearance	Erosion to one side of the ground electrode.
Cause	Normal wear in response to severe duty cycle applications.
Action	Measure the gap. Replace the spark plug if the gap is out of specification.



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Figure 1A



13300066

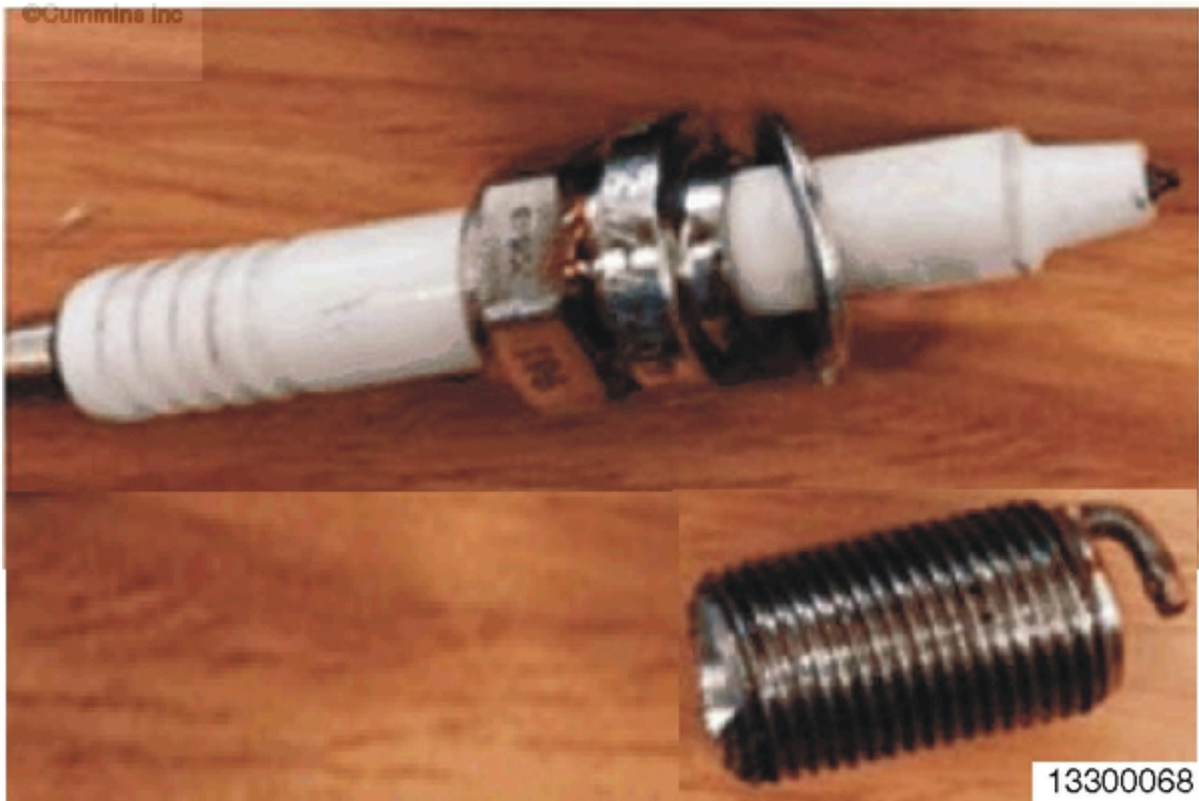
Figure 2A



Figure 2B

Excessive Dielectric Grease on Spark Plug

Appearance	Excessive dielectric grease on the spark plug terminal or insulator. Figure 1 - Normal, Figure 2A and 2B - excessive dielectric grease.
Cause	Use of additional dielectric grease. Grease buildup in the ignition coil.
Action	Compare with spark plugs from the other cylinders to determine if the issue is isolated to one cylinder or across all cylinders. Remove the grease deposits and reuse spark plug. Replace the old boot with new, pre-greased, boot when replacing spark plug.



Spark Plug - Threads Detached

Appearance	Spark plug detached from the threads.
Cause	Over-torqued.
Action	Replace the spark plug.



13300069

Spark Plug - Melted Terminal

Appearance	Top of the spark plug is melted.
Cause	Excessive temperatures or knock.
Action	Compare with the other spark plugs. Check for knock and/or possible in-cylinder damage. Replace all the spark plugs.



13300070

Boot Stuck on Spark Plug	
Appearance	Coil boot covering the spark plug with metal shell.
Cause	Coil installed with the boot from the previous coil still on spark plug, Normal wear.
Action	Replace all the spark plugs with damaged boots.



Figure 1



Figure 2

Wrong Boot on Coil, Causing Short Spark Plug Life - ISL G

Appearance

Larger, orange boot. Figure 1 - Wrong boot.
Figure 2 - Correct boot.

Cause

Spark plug kit ordered for the C Gas engine.

Wrong Boot on Coil, Causing Short Spark Plug Life - ISL G	
Action	Replace with the spark plug kit for ISL G.

Document History

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